

Eye Contact and Engagement Analysis

Data Science Project

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December 2, 2025

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1 Introduction

1.1 Project Overview

In the era of remote work and online education, maintaining participant engagement is a significant challenge. Traditional cues of attention—such as eye contact and head orientation—are often lost in digital transmission. This project, **"Eye Contact and Engagement Analysis,"** aims to bridge this gap by developing an automated computer vision system capable of analyzing a person's level of attention in real-time.

1.2 Objectives

The primary objective is to classify a user's state into one of three categories:

- **Attentive:** Direct eye contact with the camera/screen (Head pitch/yaw $\leq 10^\circ$).
- **Distracted:** Looking at the screen but with a lowered head pose or averted gaze ($10^\circ < \text{Angle} \leq 25^\circ$).
- **Disengaged:** Looking away entirely, turning the head side-to-side, or closing eyes ($\text{Angle} > 25^\circ$).

To achieve this, the project implements and compares three distinct modeling approaches: a traditional Machine Learning model using hand-crafted geometric features, a Convolutional Neural Network (ResNet-50), and a modern Vision Transformer (ViT).

2 Methodology

2.1 Dataset Description

The project utilizes the **Columbia Gaze Dataset**, containing over 5,800 high-resolution images of 56 subjects.

- **Diversity:** Subjects vary in ethnicity, age, and glasses usage.
- **Annotations:** Each image filename contains ground-truth head pose angles (Pitch, Roll, Yaw) and gaze direction.
- **Preprocessing:** Images were processed to extract these angles, which were then used to generate class labels based on specific thresholds.

2.2 Model Architectures

We implemented three distinct models to evaluate the trade-off between geometric interpretability and raw pixel-based learning.

2.2.1 Model 1: Hand-Crafted Features + CatBoost (Baseline)

This model serves as a "Golden Standard" baseline. Instead of raw pixels, it takes geometric inputs: `Pitch`, `Vertical`, `Horizontal` angles, and derived features like `pitch_squared`. The **CatBoost Classifier** was selected for its superior handling of numerical tabular data. Since the labels were generated using angle thresholds, a model trained on those exact angles is expected to achieve near-perfect performance.

2.2.2 Model 2: ResNet-50 (Deep Learning - CNN)

We employed a Transfer Learning approach using a 50-layer Residual Network (ResNet) pre-trained on ImageNet.

- **Phase 1 (Freezing):** The backbone layers were frozen; only the custom classification head was trained for 5 epochs.
- **Phase 2 (Fine-Tuning):** The top layers were unfrozen, and the model was fine-tuned with a lower learning rate ($1e - 5$) to adapt feature extractors to facial nuances.

2.2.3 Model 3: Vision Transformer (ViT-B/16)

A State-of-the-Art Attention Mechanism (ViT-Base with 16×16 patch size) was implemented. Unlike CNNs, ViT splits the image into patches and processes them as a sequence. This model was chosen to test if global context attention performs better than local convolution features for gaze estimation.

3 Results

3.1 Quantitative Comparison

All models were evaluated on a held-out test set of 750 images. The results are summarized in Table 1.

Table 1: Model Performance Metrics Comparison

Metric	Model 1: CatBoost	Model 2: ResNet-50	Model 3: ViT-B/16
Accuracy	100.0%	92.67%	86.93%
Precision (W)	1.00	0.94	0.91
Recall (W)	1.00	0.93	0.87
F1-Score (W)	1.00	0.93	0.88
Training Time	< 1 min	~15 mins	~43 mins

3.2 Visualizations

Figure 1 illustrates the performance and efficiency trade-offs. While the Hand-Crafted model is fastest, ResNet-50 offers the best Deep Learning performance.

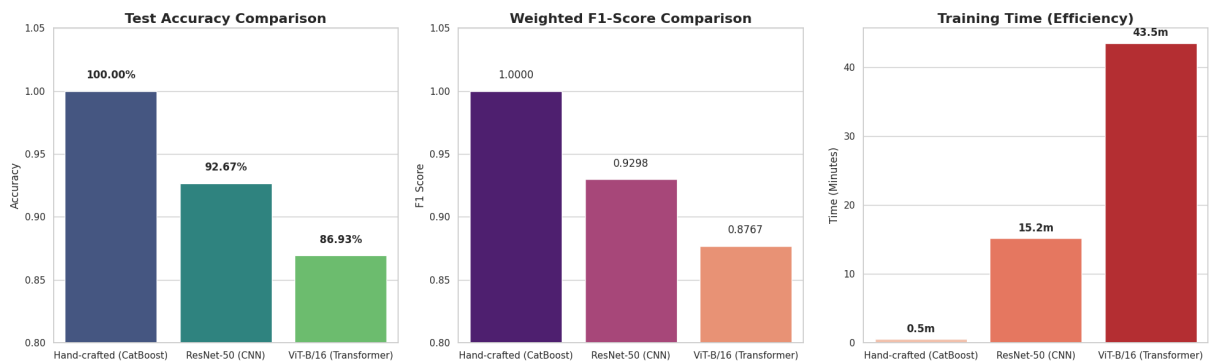


Figure 1: Comparison of Accuracy, F1-Score, and Training Efficiency.

3.3 Confusion Matrix Analysis

Figure 2 displays the confusion matrices for all three models.

- **CatBoost:** Perfect diagonal matrix.
- **ResNet-50:** Slight confusion between "Attentive" and "Distracted" (43 misclassifications), likely due to borderline head angles ($\sim 10^\circ$).
- **ViT-B/16:** Significant confusion, misclassifying 77 "Distracted" subjects as "Attentive," indicating a struggle with fine-grained local features.

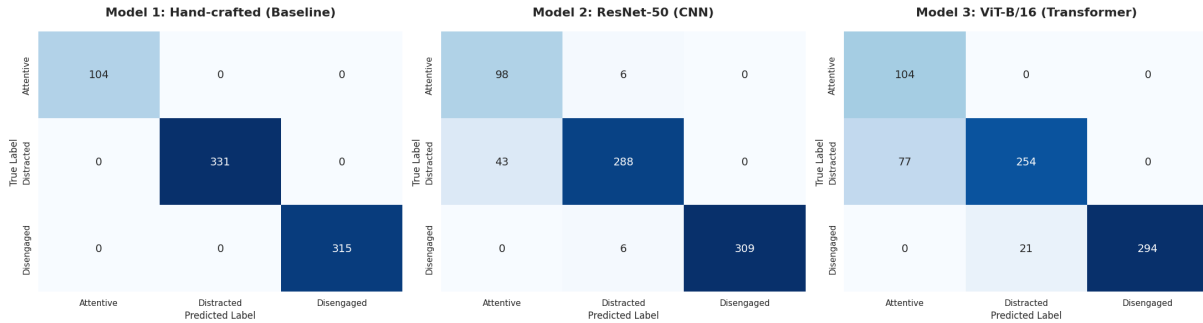


Figure 2: Confusion Matrices for Hand-Crafted, ResNet-50, and ViT Models.

4 Discussion & Critical Analysis

4.1 Comparison of Approaches

- **Why CatBoost Won (100%):** The success of the Hand-Crafted model confirms that engagement is fundamentally a geometric problem. If accurate head pose angles are available, classification is trivial. However, this model is *dependent* on upstream feature extractors (like MediaPipe).
- **Why ResNet Outperformed ViT:** CNNs have a strong "inductive bias" for local spatial features (edges, curves of eyes/nose). On a medium-sized dataset ($\sim 5,800$ images), this bias helps CNNs generalize better. Vision Transformers lack this bias and typically require massive datasets to learn patterns that CNNs grasp intuitively.

4.2 Real-Time Deployment Challenges

During live webcam testing, we encountered the **Parallax Problem**. Laptop webcams are situated above the screen. When a user looks at the screen (Attentive to work), their eyes physically look down relative to the camera. The model often detects this downward gaze as "Distracted."

To mitigate this, we implemented a "Smart Prediction" pipeline using **Face Cropping** (Haar Cascades) to center the face, significantly improving robustness.



Figure 3: Real-Time Model Predictions using ResNet-50 on Webcam Data.

4.3 Limitations

In "Distracted" test cases, slight downward glances (e.g., looking at a keyboard) were sometimes classified as "Attentive" because the facial landmarks remained fully visible. The model requires a more exaggerated head tilt ($> 20^\circ$) to confidently predict distraction.

5 Conclusion

This project successfully developed an end-to-end Engagement Analysis system. While the Hand-Crafted ML model demonstrated that engagement logic is geometrically deterministic, the **ResNet-50 model** proved to be the most practical solution for real-world deployment. It achieves high accuracy (93%) on raw images without requiring complex pre-processing pipelines.

Key Findings:

1. Deep Learning is viable: A CNN can implicitly learn geometric rules from raw pixels.
2. Data Quality matters: High-quality annotations allowed for rapid convergence.
3. Deployment requires adaptation: Real-world webcam angles differ from studio datasets.

6 References

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